

A Cross-Field Template for Reproducible Community Science

Equations, chemical compounds, code, wide figures and exact artifact references

GenesisL1 Community Publication Templates

Blockchain, computer science, chemistry, biochemistry and biology

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COMMUNITY REVIEW CROSS-FIELD PROFILE GL1-CPT-PAPER-3.1

Abstract

Background / Purpose. Reproducible community science requires a manuscript format that can express mathematical models, chemical identities, biological nomenclature, software, and verifiable artifacts without allowing infrastructure language to overwhelm the science.

Methods. A two-column $\text{\LaTeX}$ class was implemented with structured abstracts, numbered equations and compounds, language-aware code boxes, wide floats, numerical citations, and optional content-addressed metadata.

Results. The example compiled with standard $\text{\pdfLaTeX}$ and produced stable references to [equation \(1\)](#), [scheme 1](#), [listing 1](#), and [figure 3](#).

Conclusion. The resulting format supports cross-field scientific writing while keeping interpretation, limitations, provenance, and future infrastructure distinct.

Keywords: community science; reproducibility; chemistry; software; provenance

1 Introduction

Scientific manuscripts frequently connect mathematical analysis, molecular structures, biological observations, software, and computational provenance. The template treats these as compatible components rather than as reasons to impose one rhetorical model on every author.

2 Equations

Numbered equations use standard $\text{\LaTeX}$ semantics. For example, an elementary rate model can be written as

$$v = \frac{V_{\max} [S]}{K_m + [S]}, \quad (1)$$

and cited as [equation \(1\)](#). The equation remains human-readable in source and professionally typeset in the document.

3 Wide tables

The principal artifacts are summarized in [table 1](#).

4 Chemistry and biochemistry

Chemical formulae use `mhchem`, for example



Biological names may be written as *Escherichia coli*, genes as *lacZ*, and proteins as *LacZ*. These lightweight commands can be adapted to the nomenclature rules of a specific discipline.

5 Computer science and blockchain

Code is typeset with the standard `listings` package, so the default build does not require shell escape.

```
1 from hashlib import sha256
2
3 def commit(payload: bytes) -> str:
4     # Return a lowercase SHA-256 commitment.
5     return sha256(payload).hexdigest()
```

Listing 1. A minimal deterministic transformation

A Solidity fragment can be cited independently:

```
1 pragma solidity ^0.8.24;
2
3 contract ArtifactCommitment {
4     mapping(bytes32 => address) public author;
5
6     function register(bytes32 digest) external {
7         require(author[digest] == address(0), "already
8             registered");
9         author[digest] = msg.sender;
10    }
```

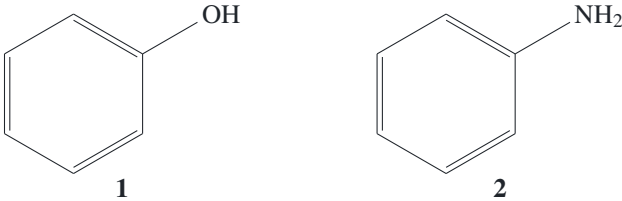
Listing 2. A content commitment stored by a contract

The result in [listing 1](#) describes a byte commitment. It does not establish the quality, legality, safety, or scientific interpretation of the committed object.

Nam dui ligula, fringilla a, euismod sodales, sollicitudin vel, wisi. Morbi auctor lorem non justo. Nam lacus libero, pretium at, lobortis vitae, ultricies et, tellus. Donec aliquet,

Table 1. Summary of experimental and computational artifacts.

Object	Description	Version	Status
Dataset	Experimental measurements used for model fitting	1.0	Released
Model	Fixed-point predictive model	1.2	Validated
Software	Reproduction pipeline	3.0	Released
Execution	Reference computational run	2026-08	Reproduced



Scheme 1. Example structures with persistent compound numbering. Compounds **1** and **2** can be cited consistently throughout the manuscript.

tortor sed accumsan bibendum, erat ligula aliquet magna, vitae ornare odio metus a mi. Morbi ac orci et nisl hendrerit mollis. Suspendisse ut massa. Cras nec ante. Pellentesque a nulla. Cum sociis natoque penatibus et magnis dis parturient montes, nascetur ridiculus mus. Aliquam tincidunt urna. Nulla ullamcorper vestibulum turpis. Pellentesque cursus luctus mauris.

6 Results

The compiled example preserved the order and labels of equations, schemes, code listings, and wide figures. All references were generated by `cleveref`.



Figure 1. An example figure caption – the image is from the *GenesisL1* press cover gallery.[1]

EVIDENCE / OBSERVATION

The build produced a two-column paper in which the official GenesisL1 lockup appeared only in the first-page front matter. Subsequent page headers contained text only.

Nam dui ligula, fringilla a, euismod sodales, sollicitudin vel, wisi. Morbi auctor lorem non justo. Nam lacus libero, pretium at, lobortis vitae, ultricies et, tellus. Donec aliquet, tortor sed accumsan bibendum, erat ligula aliquet magna, vitae ornare odio metus a mi. Morbi ac orci et nisl hendrerit mollis. Suspendisse ut massa. Cras nec ante. Pellentesque a nulla. Cum sociis natoque penatibus et magnis dis parturient montes, nascetur ridiculus mus. Aliquam tincidunt urna. Nulla ullamcorper vestibulum turpis. Pellentesque cursus luctus mauris.

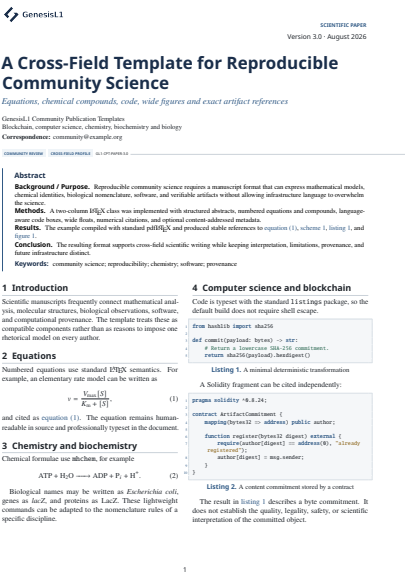


Figure 2. An example image of this paper

Item	Quantity
Candles	4
Fork handles	?

Table 2. An example table.

7 Wide figures

Single-column figures use the ordinary figure environment. Important diagrams may use `figure*` or the helper environment below to span both columns. [figure 3](#)

As illustrated in [figure 3](#), the workflow separates the scientific result from its interpretation.

The principal artifacts are summarized in [table 1](#).

Mathematics

$\LaTeX$ is great at typesetting mathematics. Let $X_1, X_2, \dots, X_n$ be a sequence of independent and identically distributed random variables with $E[X_i] = \mu$ and $\text{Var}[X_i] = \sigma^2 < \infty$,

Scientific question → **exact inputs** → **method or model** → **result** → **interpretation**

Each arrow denotes an explicit dependency, not automatic scientific validation.

Figure 3. A full-width figure spanning the two-column scientific body.

and let

$$S_n = \frac{X_1 + X_2 + \cdots + X_n}{n} = \frac{1}{n} \sum_i^n X_i$$

denote their mean. Then as n approaches infinity, the random variables $\sqrt{n}(S_n - \mu)$ converge in distribution to a normal $\mathcal{N}(0, \sigma^2)$.

Citations

LaTeX formats citations and references automatically using the bibliography records in your .bib file, which you can edit via the project menu. Use the cite command for an inline citation, like [3], and the citep command for a citation in parentheses [2].

Lists

You can make lists with automatic numbering ...

1. Like this,
2. and like this.

... or bullet points ...

- ▶ Like this,
- ▶ and like this.
- ▶ or cite a compound 2

... or with words and descriptions ...

Word Definition

Concept Explanation

Idea Text

8 Discussion

Our results provide evidence that a single paper class can support the selected fields without making every manuscript about blockchain infrastructure. This observation points towards a broader community publication system, but suitability for a specific journal or reporting guideline must still be evaluated separately.

Interpretation

The main value is not visual uniformity by itself. It is the consistent distinction among direct observations, interpretations, limitations, and exact supporting artifacts.

Limitations and uncertainty

This example is a template demonstration rather than an empirical validation study. Chemical structures generated from SMILES are intentionally left to a

separate, reviewable preprocessing step. The default manuscript build consumes committed graphics or chemfig source and does not execute external converters.

9 Conclusion

A supportive manuscript container can provide technical precision while leaving scientific judgement, disciplinary conventions, and authorial voice intact.

Artifact record

Provenance and reproducibility

Data	doi:10.xxxx/example-dataset
Code	git:0123456789abcdef
Model	model:v1.2.0
Execution	run:example-2026-08-01
Content hash	sha256:placeholder
IPFS CID	bafy...placeholder
Gateway hint	Optional resolver supplied by the author

Template principle. A supportive container, not a Procrustean bed. Remove modules that do not serve the paper and add field-specific reporting requirements where appropriate.

References

- [1] GenesisL1. A press release form. <https://genesisl1.com/press/>, 2026. Press stuff, Version 1.0.
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